Unit Test Report

Project: Student Attendance Management System. Activity: Part 3 testing of completed Sprint 1 and Sprint 2 work. Execution dates: 6 November 2026 (Part 3 run, branch `feature/testing-feedback`, base commit `438845cb25f6`) and 9 November 2026 (full regression on `main` after the timetable, semester classes, send-login and interface changes, BL-01 D-24 to D-28). No Sprint 3 was started.

Result

The final Django suite completed successfully: **552 passed, 4 skipped, zero failures and zero errors**, out of 556 discovered tests. All executed tests passed. Skips remain unexecuted checks, not passes.

Execution started 09 November 2026 at 16:35:02 IST and ended 16:36:32 IST. Django reported 89.939 seconds for the suite. The 6 November run (526 passed, 4 skipped, 530 tests) is superseded by this rerun. The temporary test database was created and destroyed successfully.

Scope and method

This required Unit Test Report covers the repository's automated suite, including unit/component assertions, database-backed tests, request-level tests, concurrency tests and 15 cross-module integration journeys. The total must not be described as 556 isolated unit tests. The System Validation Report separates browser and performance evidence.

The working tree adds two validation tests, 30 coverage follow-up tests in five new test files, strengthens one browser assertion, and pins coverage.py 7.17.2. The application code was unchanged on 6 November; the 9 November rerun includes the application changes recorded in BL-01 D-24 to D-28. Results and coverage in this report come from the final full rerun after the follow-up tests were added; an earlier run on the same day measured 92.89% line and 85.38% branch coverage before them. Coverage measures the automated Django run only; browser and performance results are separate evidence.

The opt-in `verification.test_runner.Runner` records each test name, outcome and duration. Coverage runs with branch instrumentation enabled. The project runner disables OTP for older non-OTP tests; OTP and real-login integration tests enable it explicitly. Some OTP, credential-email and backup test classes explicitly use the MD5 test hasher for speed; this does not change application password storage, and is not a production hashing-strength measurement.

Results by area

Area	Passed	Skipped	Failed or error
apps.academics	116	0	0
apps.accounts	162	1	0
apps.attendance	148	0	0
apps.audit	6	0	0
apps.backups	41	3	0
apps.notifications	32	0	0
config	5	0	0
tests.integration	15	0	0
tests.other	27	0	0

New verification assertions

- UT_02_3 now checks that a malformed student email rejects the form, creates neither a user nor a profile, and sends no credentials.

- UT_05_2 now checks both required section fields together and individually, while preserving the optional batch-year behavior.

- UT_06_6 is checked in the browser matrix: filling the remaining marks preserves an explicitly selected absence and the saved record.

The two added Django methods passed in this final full-suite run. The browser assertion passed in every final browser case. No application behavior was changed to obtain these results.

Tests added on 9 November 2026

The 9 November changes added 26 tests in five files, all passing, catalogued as UT_01_15-17, IT_01_9, IT_11_4, IT_12_5-6, UT_14_1-7 and IT_14_1-3. Two existing tests were updated to the agreed new behaviour (the dashboard shortage box is hidden when nobody is below the threshold; history lists records after the filters are applied).

Behavior checked	Test file	Tests
Sidebar shows only the role's own pages and marks the current one; sign-in pages have no sidebar	apps/accounts/test_layout.py	6
Send login details: current password only, address saved, invalid address, Admin only	apps/accounts/test_send_credentials.py	4
Timetable add/remove and clashes; semester totals and invalid values; Admin-only pages	apps/academics/test_planning.py	5
Classes-needed calculation and advice; today's classes; student dashboard, class-by-class list, notifications; dashboard count with a deactivated student	apps/attendance/test_outlook.py	10
History lists records only after filtering	apps/attendance/test_history.py	1

Coverage follow-up tests

After the first measured run, 30 tests were added for error and repeat-request paths that no test had exercised. All 30 passed in the final run. They are catalogued as UT_01_14, IT_01_8, UT_02_4, UT_02_5, ST_02_8, IT_05_7, UT_06_7, UT_07_6, ST_07_6, UT_13_4, UT_13_5 and UT_12_5.

Behavior checked	Test file	Tests
Invalid edits save nothing; repeated deactivate/reactivate is not logged twice; delete asks first; unsupported login-identifier characters; duplicates saved concurrently by another Admin	apps/academics/test_error_paths.py	10
Marking and correction refuse signed-out or deleted accounts, future dates, non-text reasons, empty classes and missing records; correction-window page errors	apps/attendance/test_error_paths.py	8
Damaged or tampered snapshots and snapshots without an active Admin are rejected; scheduler command when busy or failing	apps/backups/test_error_paths.py	6
Two-factor pages opened out of order, a lock after the password step, and account changes during setup	apps/accounts/test_otp_error_paths.py	5
The WSGI and ASGI server entry points load	config/test_entrypoints.py	1

Two kinds of test are not page-level checks. The blank-name guards cannot be reached through the web forms, because Django rejects empty input first, so their clean methods are tested directly. Concurrent duplicates, database failures, busy/failed scheduler ticks and setup races cannot be produced on demand, so those tests simulate the error the database or service would report.

Failures during execution

One automated test failed during the 9 November regression: UT_01_17 failed when the random temporary password contained "&", because the test searched for unescaped text while the page correctly showed `&`. The test was fixed (F-02, commit 3999bbd), passed six repeated runs, and the final full run above has zero failures. The other fixes found during testing (F-01, F-03, F-04, F-05) are in the System Validation Report and Test Plan Section 7.

Skipped checks

Check	Reason and impact
test_otp_is_required_by_default	OTP_REQUIRED is set in this local environment
test_windows_refuses_a_junction_in_a_backup_path	Windows reparse-point handling is verified on Windows.
test_windows_rejects_an_existing_public_directory_acl	Windows access control is verified on Windows.
test_windows_rejects_public_file_acl_using_the_opened_handle	Windows access control is verified on Windows.

The configured environment explicitly sets OTP_REQUIRED=True, so the test for an unspecified environment default skips. Functional OTP tests still execute with OTP enabled. Three storage checks require native Windows ACL/reparse-point behavior and cannot execute on this Mac. Historical Windows evidence is retained elsewhere in Documents/Verification, but is not counted as fresh current-revision execution.

Environment and supporting checks

Python 3.14.6; Django 5.2.18; PostgreSQL 18.4; coverage.py 7.17.2; macOS 26.6.2 arm64 with 10 logical CPUs. Lint, format, Django system check and migration check were rerun on 9 November 2026 and pass. Email uses the in-memory test backend. Application data and external mailboxes were not used.

Check	Result
lint	PASS
format	PASS
django-check	PASS
migrations	PASS

Evidence and reproduction

Results of the final rerun (test outcomes, test log, coverage JSON and the HTML coverage pages) are kept locally in `artifacts/part3/rerun-2026-11-09/`. Logs and sources from the first run are retained in a separate local supporting-files archive. Neither is included in this submission folder.

The exact method results are in regression JSON and named test outcomes. Commands and exit codes are in execution commands; the console result is in test log. The updated case catalogue maps case IDs to these methods.

Run with the pinned development dependencies, an explicit loopback disposable DATABASE_URL, a test-only SECRET_KEY, OTP_REQUIRED=True and captured email. From the repository root:

```
OTP_REQUIRED=True VERIFICATION_RESULTS=artifacts/part3/rerun-2026-11-09/regression.json \
  .venv/bin/python -m coverage run --data-file=artifacts/part3/rerun-2026-11-09/.coverage \
  manage.py test --noinput --testrunner verification.test_runner.Runner
.venv/bin/python -m coverage json --data-file=artifacts/part3/rerun-2026-11-09/.coverage \
  -o artifacts/part3/rerun-2026-11-09/coverage.json
.venv/bin/python -m coverage html --data-file=artifacts/part3/rerun-2026-11-09/.coverage \
  -d artifacts/part3/rerun-2026-11-09/html
```